

Statistical Tests on the Documented Record

What the data published at palestinerecord.github.io shows, tested

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Summary

This report sets out 12 hypothesis tests on the data the dashboard publishes, each with its null hypothesis, its data, its test, its effect size and its confidence interval. Of the 11 that produce a p-value, 8 reject their null hypothesis after correction for multiple testing; the three that do not are the ICJ orders, after which no fall in the death rate can be detected.

Test	Question	Result	Effect
H1b	Age and sex of the dead: pattern confined to men of fighting age	null rejected	Share of dead who are women, children under 18 or aged 60+: 0.521 (95% CI 0.517 to 0.525)
H2a	The November 2023 pause and the daily killing rate	null rejected	Rate ratio, pause against the 14 days before: 0.0025 (95% CI 0.00135 to 0.00468)
H2b	The January-March 2025 ceasefire and the daily killing rate	null rejected	Rate ratio, ceasefire against the 60 days before: 0.149 (95% CI 0.0506 to 0.44)
H2c	The October 2025 ceasefire and the daily killing rate	null rejected	Rate ratio, ceasefire against the 60 days before: 0.039 (95% CI 0.0274 to 0.0553)
H3a	The ICJ order of 26 January 2024 and the daily killing rate	null not rejected	Rate ratio, level change at the order: 1.01 (95% CI 0.808 to 1.25)
H3b	The ICJ order of 28 March 2024 and the daily killing rate	null not rejected	Rate ratio, level change at the order: 0.819 (95% CI 0.645 to 1.04)
H3c	The ICJ order of 24 May 2024 and the daily killing rate	null not rejected	Rate ratio, level change at the order: 1.11 (95% CI 0.703 to 1.77)
H4a	West Bank killings before and after 7 October 2023	null rejected	Rate ratio, after against before: 4.66 (95% CI 3.89 to 5.59)
H4b	Settler incidents against Palestinians, 2024 against 2022	null rejected	Rate ratio, 2024 against 2022: 3.5 (95% CI 3.22 to 3.81)
H6a	The identity numbers on the named list	validity not rejected; the 10% fabrication benchmark is rejected	Share of identity numbers with a valid check digit: 1 (95% CI 0.99995 to 1)
H6b	Age heaping on the named list	highly accurate	Whipple's index: 100
H8b	Recognition of Palestine and calling it genocide	null rejected	Odds ratio: 34.3 (95% CI 2.06 to 570)

In plain terms, the findings are these.

- **The named list of the dead is genuine on every internal test.** Every one of the 72,835 identity numbers on the Ministry of Health list passes the Population Registry check digit, which a fabricated number passes about one time in ten; none is duplicated; and every stated age agrees with its date of birth.
- **The dead are not confined to men of fighting age.** Just over half of the identified dead are women, children or people aged 60 and over.
- **Every ceasefire cut the rate of killing, and none stopped it.** Killings continued through the November 2023 pause, the January to March 2025 ceasefire and the ceasefire in force since October 2025.

- The three ICJ provisional measures orders of 2024 were not followed by any detectable fall in the daily death rate.
- In the West Bank, Palestinians were killed at more than four times the earlier rate after 7 October 2023, and settler incidents in 2024 ran at about three and a half times the 2022 rate.
- No state that withholds recognition of Palestine has a government that has called the conduct in Gaza genocide; 50 recognising states do.

Method

Tests. Counts are modelled as negative binomial, after a likelihood-ratio test rejected the Poisson model in every series where it was applied. Time series use Newey-West standard errors with a lag of seven days. Rates are compared with the exact conditional test for two Poisson rates. Two-by-two tables use Fisher's exact test.

Multiple testing. p-values are adjusted for the false discovery rate (Benjamini-Hochberg, 5%) across all 15 tests run in the analysis. Both raw and adjusted p-values are reported.

Caution. A change in a series at a dated event is an association, not proof that the event caused it. Tests on the named list inherit any bias in who has been identified.

What the tests are not. A test of a ceasefire or a court order measures whether the death rate changed around a date, not why. A test on the named list measures the list as published, not the deaths it does not contain: the 1,205 counted dead who have not been identified, and the dead still under the rubble, are outside it.

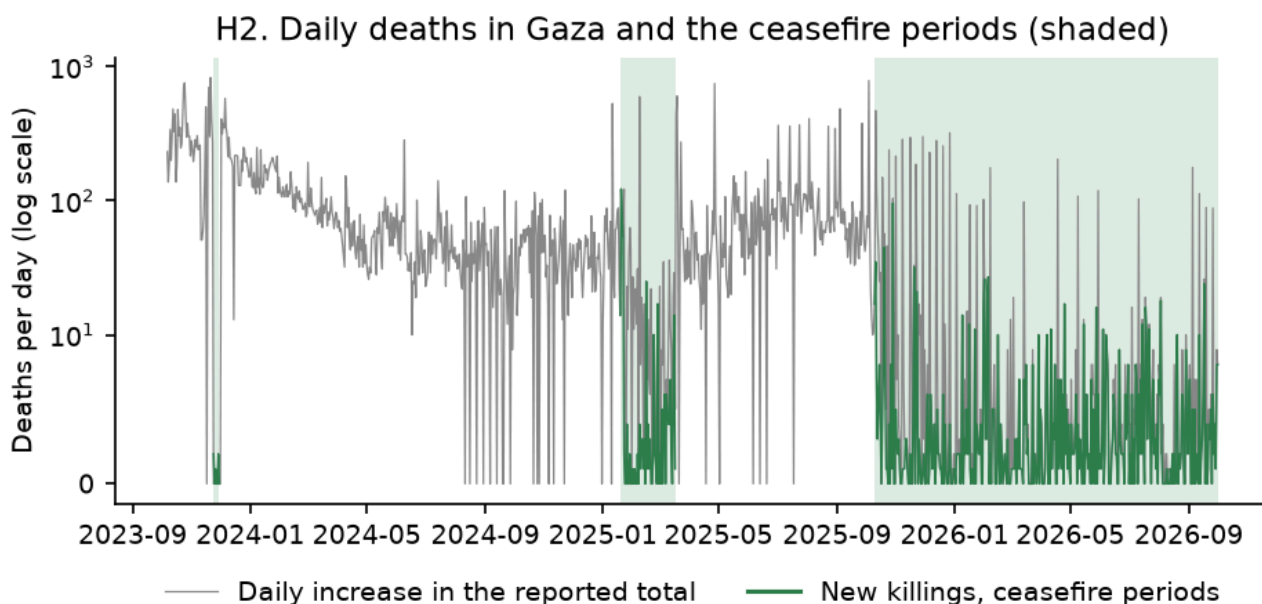
H1. Who is killed: age and sex against the population

H1b. Age and sex of the dead: pattern confined to men of fighting age

52.1% of the identified dead (95% CI 51.7% to 52.5%) are women, children under 18 or people aged 60 and over, against 75.5% of the population. More than half of the dead fall outside the group from which armed men are drawn: the dead are not confined to men of fighting age.

Item	Detail
Null hypothesis	The identified dead are confined to men aged 18-59.
Data	Ministry of Health named list, 72,835 identified dead; PCBS/UNFPA Gaza Strip population 2023 (2,226,544).
Test	Exact binomial test of the share of women, children and people aged 60 and over against zero
Effect	Share of dead who are women, children under 18 or aged 60+: 0.521 (95% CI 0.517 to 0.525)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Main report	§6.1

H2. The ceasefires



H2a. The November 2023 pause and the daily killing rate

Reported killings fell from 284.4 a day in the fortnight before the pause to 0.71 a day during it (rate ratio 0.0025, 95% CI 0.0013 to 0.0047). Killing did not stop: at least 5 people were reported killed in the 7 days. Against the upper bound of 40 deaths the ratio is 0.0201, still a fall of about 98%.

Item	Detail
Null hypothesis	The daily rate of killing during the pause of 24-30 November 2023 equalled the rate in the 14 days before it.
Data	Comparison window: Ministry of Health / Government Media Office daily series (Tech For Palestine), 10-23 November 2023. Pause: killings reported incident by incident (OCHA Flash Updates #49-#55; AP; Al Jazeera).
Test	Negative binomial regression on a pause indicator, Newey-West errors (lag 7)
Effect	Rate ratio, pause against the 14 days before: 0.0025 (95% CI 0.00135 to 0.00468)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Sensitivity: Exact conditional Poisson test	0.0025 (95% CI 0.000815 to 0.00587), $p < 0.001$
Sensitivity: Against the upper bound of 40 deaths during the pause	0.0201 (95% CI 0.0143 to 0.0274), $p < 0.001$
Main report	§13

Sources for the pause-period count: OCHA, "Hostilities in the Gaza Strip and Israel - Flash Update #49", 24 November 2023, <https://www.ochaopt.org/content/hostilities-gaza-strip-and-israel-flash-update-49>; OCHA, "Hostilities in the Gaza Strip and Israel - Flash Update #51", 26 November 2023, <https://www.ochaopt.org/content/hostilities-gaza-strip-and-israel-flash-update-51>; OCHA, "Hostilities in the Gaza Strip and Israel - Flash Update #53", 28 November 2023, <https://www.ochaopt.org/content/hostilities-gaza-strip-and-israel-flash-update-53>; OCHA, "Hostilities in the Gaza Strip and Israel - Flash Update #55", 30 November 2023, <https://www.ochaopt.org/content/hostilities-gaza-strip-and-israel-flash-update-55>; Associated Press (PBS NewsHour), "Israeli troops fire at Palestinians attempting to return to northern Gaza during cease-fire", 24 November 2023, <https://www.pbs.org/newshour/world/israeli-troops-fire-at-palestinians-attempting-to-return-to-northern-gaza-during-cessate-fire>; Al

Jazeera, "'The war is not over': Israel blocks Palestinians' return to northern Gaza", 24 November 2023, <https://www.aljazeera.com/news/2023/11/24/displaced-people-attempt-to-return-home-across-gaza-as-truce-sets-in>.

H2b. The January-March 2025 ceasefire and the daily killing rate

New killings fell from 48.8 a day in the 60 days before to 7.28 a day during the ceasefire (rate ratio 0.149, 95% CI 0.051 to 0.44). Killing continued throughout: 422 new killings were reported in 58 days (95% CI 6.6 to 8.0 a day).

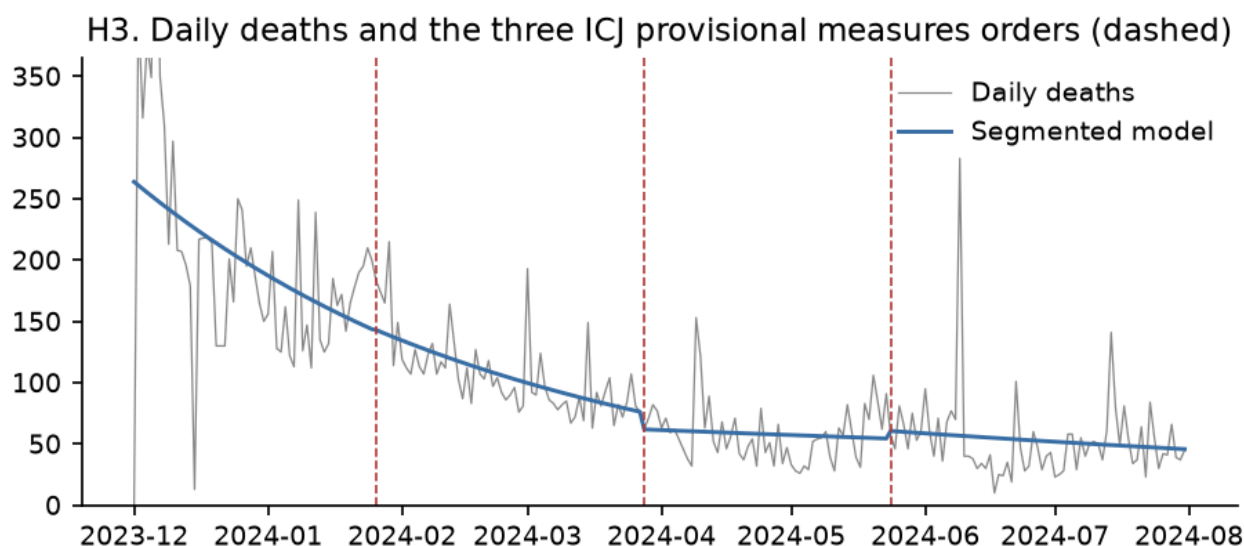
Item	Detail
Null hypothesis	The daily rate of new killings during the ceasefire (19 Jan 2025 to 17 Mar 2025) equalled the rate in the 60 days before it.
Data	Ministry of Health daily series (Tech For Palestine). Ceasefire days count new killings only where the Ministry separates them from bodies recovered.
Test	Negative binomial regression on a ceasefire indicator, Newey-West errors (lag 7)
Effect	Rate ratio, ceasefire against the 60 days before: 0.149 (95% CI 0.0506 to 0.44)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Sensitivity: Weekly totals	0.152 (95% CI 0.0654 to 0.355), p < 0.001
Sensitivity: New killings only in both windows	0.18 (95% CI 0.0652 to 0.496), p < 0.001
Main report	§13

H2c. The October 2025 ceasefire and the daily killing rate

New killings fell from 96.1 a day in the 60 days before to 3.74 a day during the ceasefire (rate ratio 0.039, 95% CI 0.027 to 0.055). Killing continued throughout: 1,336 new killings were reported in 357 days (95% CI 3.54 to 3.95 a day).

Item	Detail
Null hypothesis	The daily rate of new killings during the ceasefire (10 Oct 2025 to 01 Oct 2026) equalled the rate in the 60 days before it.
Data	Ministry of Health daily series (Tech For Palestine). Ceasefire days count new killings only where the Ministry separates them from bodies recovered.
Test	Negative binomial regression on a ceasefire indicator, Newey-West errors (lag 7)
Effect	Rate ratio, ceasefire against the 60 days before: 0.039 (95% CI 0.0274 to 0.0553)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Sensitivity: Weekly totals	0.037 (95% CI 0.0234 to 0.0576), p < 0.001
Sensitivity: New killings only in both windows	0.061 (95% CI 0.0465 to 0.08), p < 0.001
Main report	§13

H3. The ICJ provisional measures orders



H3a. The ICJ order of 26 January 2024 and the daily killing rate

No detectable change in the level or the trend of daily deaths at the order (level rate ratio 1.006, 95% CI 0.81 to 1.3; joint test $p = 0.99$). The order was not followed by a measurable fall in the rate of killing. This is the absence of a detectable change, not proof that the order had no effect on any conduct.

Item	Detail
Null hypothesis	No change in the level or the slope of daily deaths at the order of 26 January 2024.
Data	Ministry of Health daily series (Tech For Palestine), 1 December 2023 to 31 July 2024.
Test	Segmented negative binomial regression (trend, and a level and slope change at each order), Newey-West errors; joint Wald test of level and slope
Effect	Rate ratio, level change at the order: 1.01 (95% CI 0.808 to 1.25)
p (raw / adjusted)	0.989 / 0.989
Result	null not rejected
Main report	§13

H3b. The ICJ order of 28 March 2024 and the daily killing rate

No detectable change in the level or the trend of daily deaths at the order (level rate ratio 0.819, 95% CI 0.65 to 1; joint test $p = 0.16$). The order was not followed by a measurable fall in the rate of killing. This is the absence of a detectable change, not proof that the order had no effect on any conduct.

Item	Detail
Null hypothesis	No change in the level or the slope of daily deaths at the order of 28 March 2024.
Data	Ministry of Health daily series (Tech For Palestine), 1 December 2023 to 31 July 2024.
Test	Segmented negative binomial regression (trend, and a level and slope change at each order), Newey-West errors; joint Wald test of level and slope
Effect	Rate ratio, level change at the order: 0.819 (95% CI 0.645 to 1.04)

Item	Detail
p (raw / adjusted)	0.157 / 0.181
Result	null not rejected
Main report	§13

H3c. The ICJ order of 24 May 2024 and the daily killing rate

No detectable change in the level or the trend of daily deaths at the order (level rate ratio 1.115, 95% CI 0.7 to 1.8; joint test $p = 0.81$). The order was not followed by a measurable fall in the rate of killing. This is the absence of a detectable change, not proof that the order had no effect on any conduct.

Item	Detail
Null hypothesis	No change in the level or the slope of daily deaths at the order of 24 May 2024.
Data	Ministry of Health daily series (Tech For Palestine), 1 December 2023 to 31 July 2024.
Test	Segmented negative binomial regression (trend, and a level and slope change at each order), Newey-West errors; joint Wald test of level and slope
Effect	Rate ratio, level change at the order: 1.11 (95% CI 0.703 to 1.77)
p (raw / adjusted)	0.812 / 0.870
Result	null not rejected
Main report	§13

H4. The West Bank before and after 7 October 2023

H4a. West Bank killings before and after 7 October 2023

Palestinians were killed in the West Bank at 3.48 a day after 7 October 2023 against 0.75 before it, 4.66 times the rate (95% CI 3.89 to 5.59), in a year that was already the deadliest on record before the war began.

Item	Detail
Null hypothesis	The daily rate of Palestinians killed in the West Bank was the same after 7 October 2023 as before it, within 2023.
Data	OCHA via the report: 208 killed in 279 days (1 January-6 October 2023); 299 in 86 days (7 October-31 December 2023).
Test	Exact conditional test of two Poisson rates
Effect	Rate ratio, after against before: 4.66 (95% CI 3.89 to 5.59)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Main report	§11

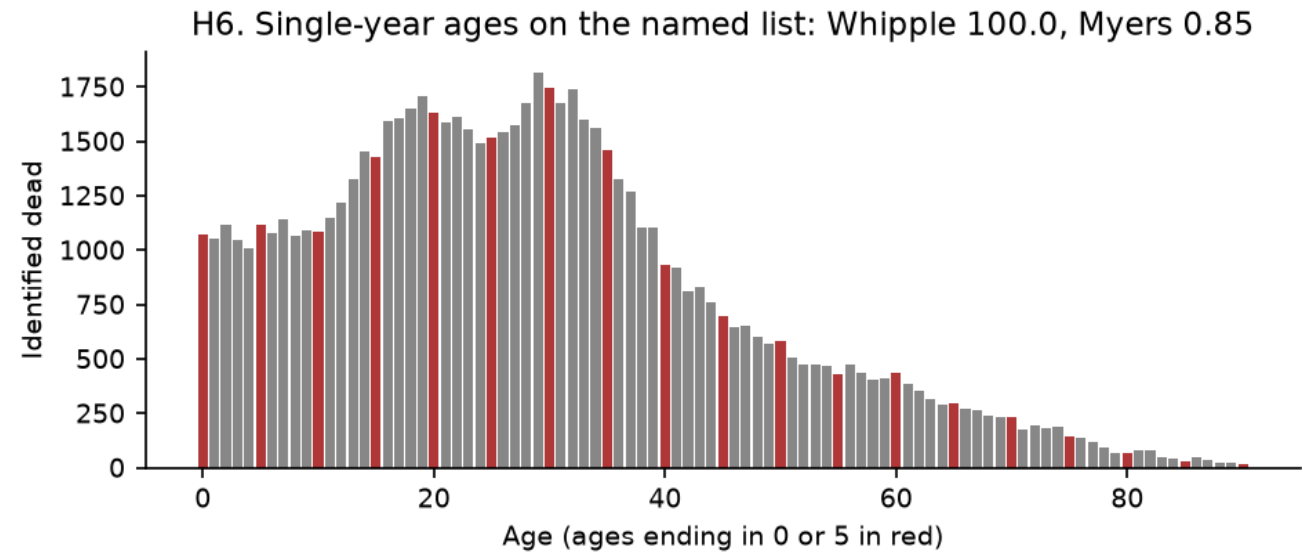
H4b. Settler incidents against Palestinians, 2024 against 2022

Settler incidents ran at 3.5 times the 2022 rate in 2024 (95% CI 3.22 to 3.81), and at 2.6 times even at the rounding bounds least favourable to the finding.

Item	Detail
Null hypothesis	The rate of settler incidents against Palestinians was the same in 2024 as in 2022.

Item	Detail
Data	OCHA daily averages via the report: 2 a day in 2022, 7 a day in 2024.
Test	Exact conditional test of two Poisson rates on annual counts reconstructed from the daily averages
Effect	Rate ratio, 2024 against 2022: 3.5 (95% CI 3.22 to 3.81)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Sensitivity: At the rounding bounds least favourable to rejection (2.5 and 6.5 a day)	2.6 (95% CI 2.41 to 2.81), p < 0.001
Main report	§11

H6. Is the named list genuine?



H6a. The identity numbers on the named list

All 72,835 of the 72,835 identity numbers pass the Population Registry check digit, where fabricated numbers would pass about one time in ten. There are 0 duplicate identity numbers and 0 duplicate name-and-birth-date pairs, and the stated age agrees with the date of birth in 72,835 of 72,835 records. The list as published is consistent with registry data. These tests cannot detect a real identity number attached to someone who did not die; that question needs matching against other records.

Item	Detail
Null hypothesis	The identity numbers are valid Population Registry numbers (pass the check digit).
Data	Ministry of Health named list, 72,835 records.
Test	Share passing the check digit, with its interval; exact binomial test against the 10% expected of fabricated numbers; decision rule: validity rejected if under 99% pass
Effect	Share of identity numbers with a valid check digit: 1 (95% CI 0.99995 to 1)
p (raw / adjusted)	< 0.001 / < 0.001
Result	validity not rejected; the 10% fabrication benchmark is rejected

Item	Detail
Main report	§6.1

H6b. Age heaping on the named list

No age heaping: Whipple's index is 100.0 and Myers' index 0.85, against the 2017 census's 100 and 2.4. Ages on the list follow from registry dates of birth rather than estimates, which is what a list built from identity records, and not from guesses, should show.

Item	Detail
Null hypothesis	Age reporting shows no more heaping on ages ending in 0 and 5 than a registry-based list would.
Data	As H6a.
Test	Whipple's index (ages 23-62) on the UN scale, and Myers' blended index (ages 10-89), against the PCBS 2017 census
Effect	Whipple's index: 100
p (raw / adjusted)	— / —
Result	highly accurate
Main report	§6.1

H8. What states have done

H8b. Recognition of Palestine and calling it genocide

50 of 157 recognising states have a government that has called it genocide; none of the 36 non-recognising states has (odds ratio 34.29, with a correction for the empty cell).

Item	Detail
Null hypothesis	Whether a UN member state government has called it genocide is independent of whether it recognises Palestine.
Data	World positions dataset, 193 UN member states.
Test	Fisher's exact test (interval with Haldane correction for a zero cell)
Effect	Odds ratio: 34.3 (95% CI 2.06 to 570)
p (raw / adjusted)	< 0.001 / < 0.001
Result	null rejected
Main report	§15.4

Notes on the data

H2a. The Ministry of Health daily series cannot be used for the pause: the Ministry had stopped reporting on 11 November, the Government Media Office issued no total during the pause, and the series records zero for every day of it, which are missing reports and not days without deaths. As directed by the editor, the pause-period count was reconstructed from the killings reported incident by incident in OCHA's flash updates and by the Associated Press and Al Jazeera (five deaths in seven days, a lower bound, since OCHA also reports shootings with 'several casualties' it does not break down). The comparison window is the 14 days before the pause, from the daily series. Because the two windows are measured differently, the test is repeated against an upper bound of 40 deaths, the rise in the Government Media Office total during the pause less the 160 bodies recovered from the rubble.

Limits

The tests are only as good as the data. The Ministry of Health's daily series changed source during the war (to the Government Media Office from 11 November 2023, and back), carries days on which no report was issued, and, from 2025, separates new killings from bodies

recovered only on some days. The named list covers the identified dead only. The population baseline is the PCBS projection for 2023, before the war. Where a test depends on one of these weaknesses, its entry above says so, and a sensitivity analysis tests whether the result survives it.

Reproducing this report

The scripts are in the source repository. They fetch the full named list from Tech For Palestine and read every other input from the dashboard's published `data/` and `data/raw/` folders.